Project Title

Brain Tumor Detection

1. Problem Statement

Paragraph 1:
Brain tumors are abnormal growths of cells inside the brain that can affect cognitive and physical functions. Early detection is critical because delayed diagnosis can increase mortality and complicate treatment. Traditional diagnosis involves examining MRI scans manually, which is time-consuming and requires expert radiologists. As demand for medical imaging increases, automated brain tumor detection systems can support healthcare professionals by improving speed, consistency, and diagnostic assistance.

Paragraph 2:
The specific problem addressed in this project is the automatic detection of brain tumors from MRI scan images using machine learning. The primary objectives are: (1) to classify MRI scans as tumor or non-tumor, (2) to assist in clinical diagnosis by providing fast and reliable predictions, and (3) to demonstrate how deep learning can be applied in medical imaging. Convolutional Neural Networks (CNNs) and transfer learning methods are used to learn visual patterns from labeled MRI images, enabling automated tumor recognition.

2. Dataset Link

Dataset Name: Brain Tumor MRI Dataset (Tumor vs No Tumor)

Source: Kaggle

Link:
https://www.kaggle.com/datasets/indk214/brain-tumor-dataset-segmentation-and-classification

Brief Description:
This dataset contains MRI images categorized into two classes: **Tumor** and **No Tumor**. The images are pre-segmented brain MRI scans captured using medical imaging techniques. The dataset is suitable for binary classification tasks and is commonly used in medical computer vision research. It is also updated recently, making it relevant for current academic work.

3. Machine Learning Models

Model 1: Convolutional Neural Network (CNN)
Purpose: To extract hierarchical spatial features from MRI images and classify them into tumor/non-tumor.
Reason for selection: CNNs are the standard approach for image-based classification tasks due to their strong feature learning capabilities and high performance in medical image analysis.

Model 2: Transfer Learning (e.g., VGG16 / ResNet50)
Purpose: To leverage pre-trained models trained on large image datasets for improved accuracy and reduced training time.
Reason for selection: Transfer learning is effective when datasets are not very large. Pre-trained models can generalize well and achieve higher accuracy with fewer epochs.

Model 3 (if applicable): Support Vector Machine (SVM)
Purpose: To serve as an optional classifier using features extracted from CNN layers.
Reason for selection: SVMs perform well on high-dimensional feature spaces and can improve classification performance in hybrid deep learning pipelines.

4. References

1. Author(s), Title, Source, Year.

2. Dataset documentation link.

3. Additional references.